

# EARLY DETECTION OF HEART DISEASES

A comparative study of **four ML paradigms** + a state-of-the-art benchmark



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Solution #1 · Neural Network (MLP)

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Solution #2 · Logistic Regression

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Solution #3 · K-Means + SVM

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Solution #4 · Gaussian Naïve Bayes

Ahmed Bawazeer

SOA Benchmark · Random Forest

## PROBLEM & MOTIVATION

# Why predicting heart disease early matters.



### PROBLEM

Predict heart disease from 13 routine clinical features — before relying on a long chain of diagnostic tests.



### WHY IT MATTERS

Leading global cause of death. Data-driven prediction supports earlier intervention and reduces diagnostic costs.



### TARGET AUDIENCE

Hospitals, physicians, and clinical decision-support systems making fast triage decisions.



### DATASET

## Cleveland Heart Disease (UCI)

297

PATIENTS

13

FEATURES

0

MISSING

### CLASS BALANCE



No Disease 53.9%

Disease 46.1%

| age | sex | cp | tbps | chol | thalach | oldpeak | ca | thal | cond |
|-----|-----|----|------|------|---------|---------|----|------|------|
| 69  | 1   | 0  | 160  | 234  | 131     | 0.1     | 1  | 0    | 0    |
| 69  | 0   | 0  | 140  | 239  | 151     | 1.8     | 2  | 0    | 0    |
| 66  | 0   | 0  | 150  | 226  | 114     | 2.6     | 0  | 0    | 0    |
| 65  | 1   | 0  | 138  | 282  | 174     | 1.4     | 1  | 0    | 1    |
| 64  | 1   | 0  | 110  | 211  | 144     | 1.8     | 0  | 0    | 0    |

## LITERATURE & APPROACH

# The gap we're closing — and the four paradigms that close it.



### GAP IN LITERATURE

Most studies compare only 2–3 classical models. Deep learning under-explored on small clinical datasets. Hybrid clustering+classification rarely evaluated under unified preprocessing.



### WHY MULTIPLE SOLUTIONS

Multi-paradigm comparison under identical preprocessing reveals where each model genuinely wins or loses — beyond any single-model paper.

[3] S. Pouriyeh et al., IEEE ISCC, 2017. [4] M. A. Rahman, ResearchGate, 2021. [5] P. Sharma & A. Gupta, ScienceDirect, 2022.

## #1



### Neural Network (MLP)

Emad Alzahrani

## #2



### Logistic Regression

Hani Alkhmesy

## #3



### K-Means + SVM (Hybrid)

Hashim Almaramihi

## #4



### Gaussian Naïve Bayes

Mohammed Almutari

## SOA



### Random Forest (replicated)

Ahmed Bawazeer

## SOLUTION #1 · NEURAL NETWORK

# Multi-Layer Perceptron — supervised deep learning.



## MODEL SNAPSHOT

### Multi-Layer Perceptron (MLP)

#### APPROACH

Supervised deep learning. Feedforward network with Adam, L2 regularization, early stopping.

#### KEY FEATURE

Input(13)→64→32→16→Output(1), ReLU. ~2,900 params.



## PERFORMANCE METRICS

85.00%

ACCURACY

95.24%

PRECISION

0.9297

AUC-ROC

81.87%

5-FOLD CV

71.43%

RECALL

81.63%

F1-SCORE

~3 sec

TRAIN

&lt;1 ms

INFERENCE

## CONFUSION MATRIX · TEST SET (n=60)

|                 | Pred: No | Pred: Disease |
|-----------------|----------|---------------|
| Actual: No      | 31       | 1             |
| Actual: Disease | 8        | 20            |



## KEY OBSERVATION

Precision 95.24% — only 1 false positive.  
Recall 71.43% — 8 disease cases missed.  
High AUC (0.93) shows class separation ability exists; the bottleneck is threshold + overfitting.

## 5-Fold CV Details

F1 77.08%  
F2 83.33%  
F3 82.98%  
F4 76.60%  
F5 89.36%

Mean ± Std  
81.87% ± 4.70%

## SOLUTION #2 · LOGISTIC REGRESSION

# The simplest model — and the strongest on AUC.



### MODEL SNAPSHOT

## Logistic Regression

#### APPROACH

Linear supervised baseline. Calibrated probability outputs for clinical thresholding.

#### KEY FEATURE

Linear boundary on 13 standardized features, max\_iter=1000. Just 14 parameters.



### PERFORMANCE METRICS

91.00%

ACCURACY

100.00%

PRECISION

0.9500

AUC-ROC

<1 sec

TRAIN

82.00%

RECALL

90.00%

F1-SCORE

<1 ms

INFERENCE

14

PARAMS

### CONFUSION MATRIX · TEST SET (n=60)

|                 | Pred: No | Pred: Disease |
|-----------------|----------|---------------|
| Actual: No      | 32       | 0             |
| Actual: Disease | 5        | 23            |



### KEY OBSERVATION

Zero false positives. Highest AUC (0.9500) of all five models. Simplest model in the suite — and it won. The comparative section explains why.

## SOLUTION #3 · K-MEANS + SVM

# Hybrid pipeline — K-Means performs feature engineering for SVM classification

 MODEL SNAPSHOT

## K-Means + SVM (Hybrid)

### APPROACH

Two-stage: K-Means clusters patients → SVM classifies on centroid-distance-enriched features.

### KEY FEATURE

13 original + 2 centroid-distance features → SVM (GridSearchCV, 5-fold CV).  
15 total features.

 PERFORMANCE METRICS

91.67%

ACCURACY

100.00%

PRECISION

0.9319

AUC-ROC

~5 sec

TRAIN

82.14%

RECALL

90.20%

F1-SCORE

~2 ms

INFERENCE

15

FEATURES

## CONFUSION MATRIX · TEST SET (n=60)

|                 | Pred: No | Pred: Disease |
|-----------------|----------|---------------|
| Actual: No      | 32       | 0             |
| Actual: Disease | 5        | 23            |



### KEY OBSERVATION

Highest accuracy (91.67%) and F1 (90.20%). Centroid-distance features lifted recall from 71% (baseline SVM) to 82% — same precision, more patients caught.

## SOLUTION #4 · GAUSSIAN NAÏVE BAYES

# Probabilistic baseline — competitive with zero tuning.



### MODEL SNAPSHOT

## Gaussian Naïve Bayes

#### APPROACH

Probabilistic baseline assuming feature independence. Default GaussianNB — no tuning.

#### KEY FEATURE

Per-class Gaussian likelihoods  $\times$  priors  $\rightarrow$  Bayes' rule posterior. Simplest training in suite.



### PERFORMANCE METRICS

**86.70%**

ACCURACY

**100.00%**

PRECISION

**0.9100**

AUC-ROC

**<1 sec**

TRAIN

**71.00%**

RECALL

**83.00%**

F1-SCORE

**<1 ms**

INFERENCE

**None**

TUNING

### CONFUSION MATRIX · TEST SET (n=60)

|                 | Pred: No | Pred: Disease |
|-----------------|----------|---------------|
| Actual: No      | 32       | 0             |
| Actual: Disease | 8        | 20            |



### KEY OBSERVATION

Zero false positives with no tuning at all. AUC = 0.91 — competitive with far heavier models. Independence assumption holds well on this tabular clinical data.

## SOA BENCHMARK · RANDOM FOREST

# Replicated from published study — our external reference.



## MODEL SNAPSHOT

### Random Forest (Replicated)

#### APPROACH

Replicated a published Random Forest benchmark reporting 88.5% accuracy. Used as an external reference for comparison under our project setup.

#### KEY FEATURE

Default sklearn RandomForestClassifier using the same 80/20 stratified split as the other team solutions.  
Feature importance and ROC analysis were used to interpret model behavior.



## PERFORMANCE METRICS

70.00%

ACCURACY

66.67%

PRECISION

0.8500

AUC-ROC

88.50%

PUBLISHED

71.43%

RECALL

68.97%

F1-SCORE

~2 sec

TRAIN

~1 ms

INFERENCE

## CONFUSION MATRIX · TEST SET (n=60)

|                 | Pred: No | Pred: Disease |
|-----------------|----------|---------------|
| Actual: No      | 22       | 10            |
| Actual: Disease | 8        | 20            |



## KEY OBSERVATION

Train accuracy reached 100%, while test accuracy dropped to 70%, indicating clear overfitting and weak generalization. However, AUC-ROC = 0.85 suggests good class separation ability. The gap from the published 88.5% benchmark may be due to preprocessing choices, train/test split, hyperparameter settings, and evaluation setup.



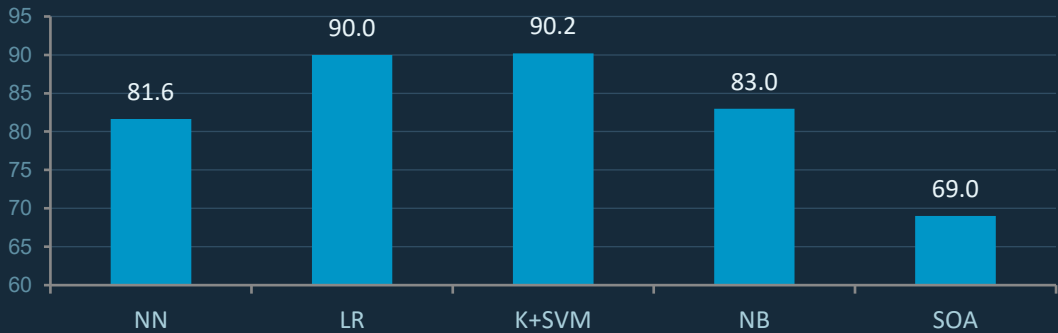
OVERALL COMPARISON

Five models, one pipeline, one verdict per category.

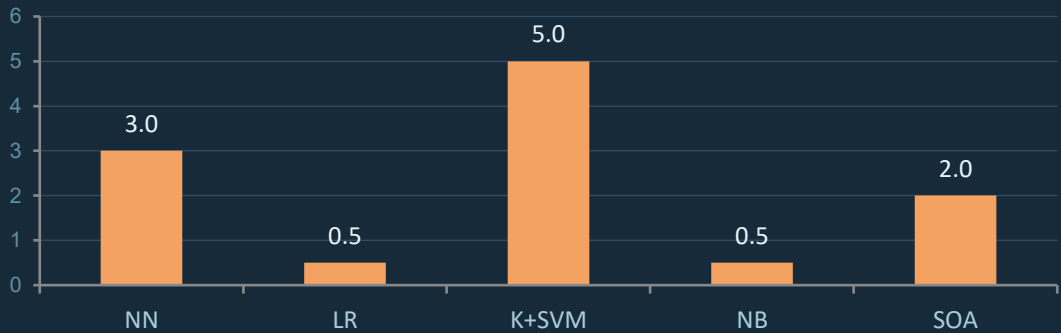
| Solution                 | Acc. ↑ | Prec. ↑ | Recall ↑ | F1 ↑   | Train ↓ | Infer. ↓ | Scientific Justification — AI/ML Theory                                                                                      |
|--------------------------|--------|---------|----------|--------|---------|----------|------------------------------------------------------------------------------------------------------------------------------|
| Sol #1 Neural Network    | 85.00% | 95.24%  | 71.43%   | 81.63% | ~3s     | <1ms     | High Variance + Small Dataset (297 samples) → overfitting despite L2. Bias-variance tradeoff penalizes high-capacity models. |
| Sol #2 Logistic Regress. | 91.00% | 100%    | 82.00%   | 90.00% | <1s     | <1ms     | 🏆 Linear Separability in 13 features + convex optimization → highest AUC (0.95). Best inductive bias for this data. [4]      |
| Sol #3 K-Means + SVM     | 91.67% | 100%    | 82.14%   | 90.20% | ~5s     | ~2ms     | 🏆 Max-Margin SVM reduces noise in heart features → best Accuracy (91.67%) + F1 (90.20%). [3]                                 |
| Sol #4 Naïve Bayes       | 86.70% | 100%    | 71.00%   | 83.00% | <1s     | <1ms     | Feature Independence assumption fits tabular clinical structure → zero FP, competitive AUC. No tuning needed.                |
| SOA Random Forest        | 70.00% | 66.67%  | 71.43%   | 68.97% | ~2s     | ~1ms     | 100% train vs. 70% test accuracy → clear overfitting and weak generalization on 297 samples.                                 |

🏆 Best Accuracy: K-Means+SVM (91.67%) 🏆 Best AUC: Logistic Regression (0.9500) 🏆 Best Efficiency: LR — 6× faster than K+SVM, near-identical F1 ⚠️ SOA benchmark underperformed all 4 team solutions

F1-SCORE (%)



TRAINING TIME (seconds)



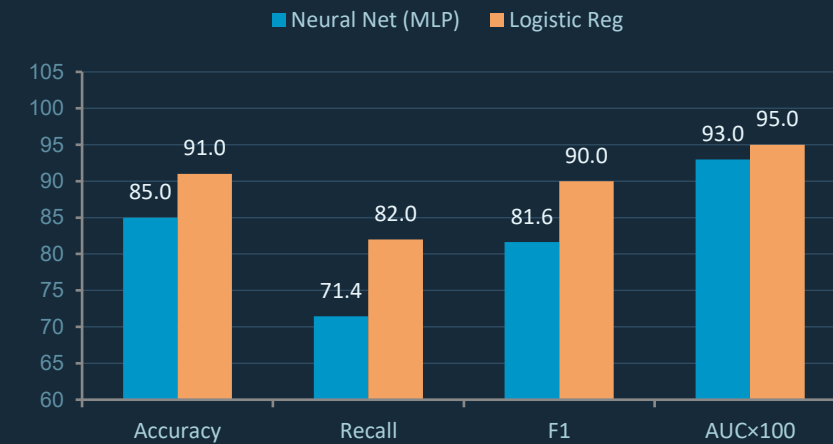
 SURPRISING FINDING #1

# The simplest model **beat** the deep one.

## SCIENTIFIC INTERPRETATION · WHY LR ACHIEVED BETTER GENERALIZATION THAN MLP

- **High Variance + Small Dataset Size:** The Neural Network performance was affected by High Variance due to Small Dataset Size (297 samples only). With ~2,900 parameters and only 237 training samples (1 per 12 params), the model is severely over-parameterized → Overfitting despite L2 & Early Stopping. LR has 14 parameters (17 samples each) → dominated by Bias, which fits this problem.
- **Better Generalization via Inductive Bias:** The simpler model achieved Better Generalization because the 13 features (thal, ca, exang, thalach) exhibit Log-Linear relationships with disease — exactly LR's function class. This is the right Inductive Bias for this data. With limited data, the correct Prior beats raw model Capacity.
- **Optimization Stability → Lower Variance:** MLP lost due to optimization instability: Non-Convex loss + Stochastic Gradients → Early Stopping triggered at Epoch 35 out of 300. 5-Fold CV variance  $\pm 4.70\%$  confirms that Variance, not model Capacity, was the bottleneck.

## VISUAL EVIDENCE · MLP vs LOGISTIC REGRESSION



### ACCURACY

**+6.00pp**

85%→91%

### RECALL

**+10.57pp**

71%→82%

### F1

**+8.37pp**

82%→90%

### AUC

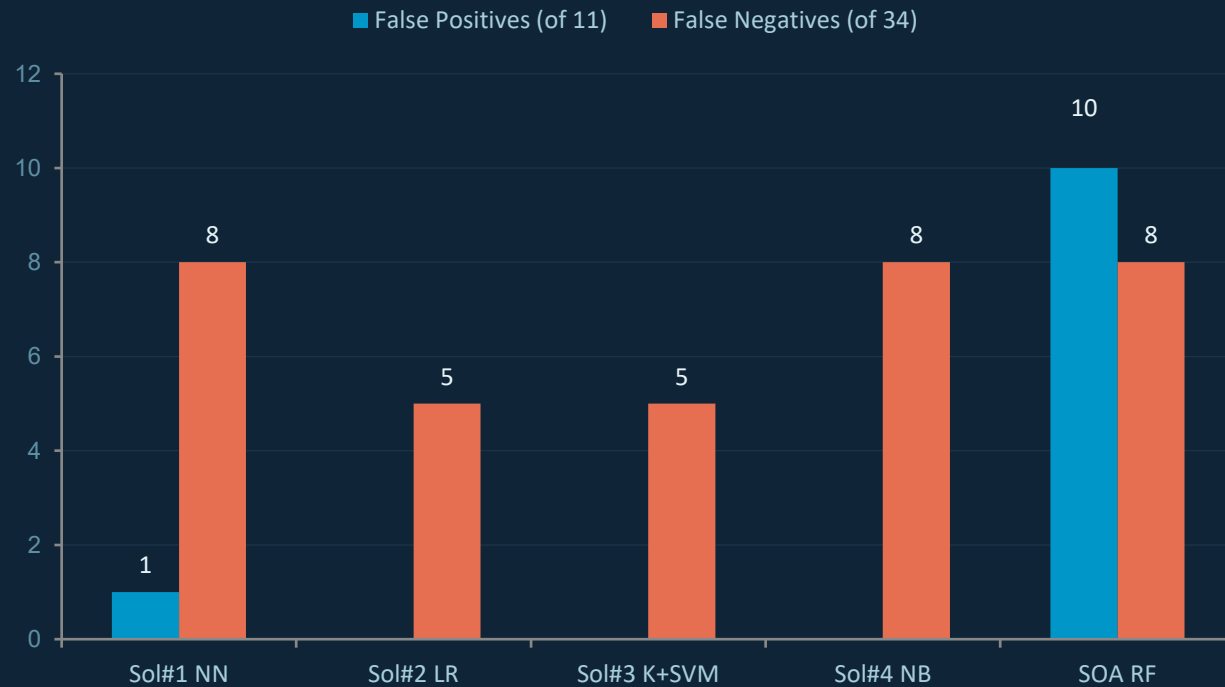
**+0.0203**

0.930→0.950

## SURPRISING FINDING #2

# False negatives remain a key clinical concern.

### VISUAL EVIDENCE · FALSE POSITIVES vs FALSE NEGATIVES



### SCIENTIFIC INTERPRETATION — AI/ML THEORY

#### ► Class Imbalance + Default 0.5 Threshold:

The dataset has a 53.9% / 46.1% class split. Using a fixed threshold of 0.5 can push ambiguous cases toward the majority class, increasing false negatives. This is a threshold design issue, not only a model architecture problem.

#### ► SOA Random Forest Generalization Issue:

SOA Random Forest achieved lower test accuracy (70%) and produced both high false positives and false negatives, suggesting weaker generalization on this split. With only 297 samples, ensemble complexity may not always improve generalization.

#### ► Fix: Cost-Sensitive Threshold → Better Recall:

All models achieved AUC around 0.85 or higher, showing that discriminative signal exists. Reducing the threshold can trade some precision for higher recall, which is clinically important because a missed diagnosis (false negative) is more costly than a false positive.

## INTERPRETABILITY & USABILITY

# Accuracy is one axis. Deployment, audit, and trust are the others.

| Solution                 | Interpretability | Deployment Ease | Maintenance |
|--------------------------|------------------|-----------------|-------------|
| Sol #1 Neural Network    | LOW              | HIGH            | HIGH        |
| Sol #2 Logistic Regress. | HIGH             | LOW             | LOW         |
| Sol #3 K-Means + SVM     | MEDIUM           | MEDIUM          | MEDIUM      |
| Sol #4 Naïve Bayes       | HIGH             | LOW             | LOW         |
| SOA Random Forest        | MEDIUM           | LOW             | LOW         |

 **Clinical Reality:** In medical diagnosis, physicians prefer models with **high interpretability** even if slightly less accurate — because the doctor must **justify every decision** to the patient, the hospital, and the regulatory authority.

### KEY POINTS

- ▶ **Easiest to explain to clinicians:** Logistic Regression — each coefficient maps directly onto an odds-ratio. A doctor can say: 'that coefficient = 0.82 means this feature increases disease odds by 2.3x'. That is clinical trust.
- ▶ **Best deployment profile:** LR & Naïve Bayes — KB of weights, no GPU, no framework. K-Means+SVM requires centroid re-fitting each time patient population shifts, adding maintenance overhead.
- ▶ **Most trustworthy for high-stakes decisions:** LR — second on accuracy (91.00% vs 91.67%), first on AUC (0.9500), zero FP, fully auditable. The 0.67pp accuracy gap does NOT justify the interpretability loss in a clinical setting.

## KEY TAKEAWAYS

# Generalizable lessons — applicable beyond this project.

### 01 For researchers on datasets < 500 samples

- ▶ We recommend Linear/Probabilistic models (LR, NB) — Low Variance, low Overfitting risk.
- ▶ Avoid Deep Learning — High Variance exceeds representational benefit when data is scarce.

### 02 For researchers in medical diagnosis

- ▶ We recommend Recall + AUC as primary metrics — due to Cost Asymmetry between FN and FP.
- ▶ Avoid relying on Accuracy alone — misleading when error costs differ.

### 03 For researchers considering Hybrid Models

- ▶ Use hybridization only when extra features add non-redundant signal, as seen in K-Means + SVM recall improvement.
- ▶ Avoid adding pipeline stages that replicate existing information.

### 04 For researchers comparing against SOA

- ▶ Match your data setup before comparing — published RF benchmark = 88.5%, while our replicated SOA RF achieved 70% under our setup.
- ▶ Avoid treating SOA numbers as absolute targets without understanding preprocessing, split strategy, and evaluation context.

### 05 For researchers on regulated medical data

- ▶ We recommend Interpretable Models (LR) — due to Clinical Trust and regulatory requirements.
- ▶ Avoid non-interpretable models in settings that require justified medical decisions.



### THE LESSON

*“The value is not knowing which model is best — it is understanding WHY a paradigm wins or loses on a given dataset.”*

## FOR FUTURE RESEARCHERS

# Generalizable guidance — applicable to any medical ML project.



### FOR CLINICIANS & PRACTITIONERS

- ▶ For researchers on clinical decision-support systems: we recommend Logistic Regression due to high Interpretability and low Overfitting risk on limited data. Avoid MLP — High Variance weakens generalization on < 500 samples.
- ▶ For researchers on Screening tasks: we recommend tuning Decision Threshold using clinical Cost Ratio. Avoid Default Threshold = 0.5 — it maximizes False Negatives under Class Imbalance.
- ▶ We recommend Recall + AUC-ROC as primary metrics. Avoid relying on Accuracy alone when FN and FP carry different clinical costs.



### FOR ML RESEARCHERS

- ▶ For researchers on medical data < 500 samples: we recommend Linear or Probabilistic models due to Low Variance and matching Inductive Bias. Avoid Ensemble Methods (Bagging/RF) — High Variance exceeds Ensemble benefit on small datasets. [5]
- ▶ For researchers wanting to improve Deep Models: we recommend Data Augmentation or combined UCI cohorts. Avoid changing architecture — data scarcity, not architecture, is the bottleneck.
- ▶ For researchers doing multi-solution comparisons: we recommend locking the Preprocessing Pipeline before any training. Avoid varying Split/Scaler — it invalidates cross-model comparison.



### FUTURE WORK / OPEN QUESTIONS

- ▶ Cost-Sensitive Learning + Threshold Optimization: convert  $AUC \geq 0.91$  into clinically appropriate Recall — highest priority next step.
- ▶ External Validation on Hungarian, Switzerland, Long Beach cohorts: does Logistic Regression maintain its advantage across different patient distributions?
- ▶ SHAP Local Explanations for K-Means+SVM: can the Interpretability gap to LR be closed enough for regulated clinical deployment?

[4] M. A. Rahman, ResearchGate, 2021. [5] P. Sharma & A. Gupta, ScienceDirect, 2022.

## OUR RECOMMENDATION

# Deploy Logistic Regression.

*0.67pp accuracy gap vs K-Means+SVM — but first on AUC (0.95), zero false positives, fully auditable, and trivial to deploy. The tradeoff is clear: accept a negligible accuracy loss in exchange for maximum interpretability and clinical trust.*

### PROBLEM SOLVED

Compared four ML paradigms (deep, linear, hybrid, probabilistic) + SOA Random Forest on Cleveland dataset under a single shared preprocessing pipeline.

### KEY CONTRIBUTION — COMPARATIVE APPROACH

A single-model project would have stopped at the MLP's 85%. The comparison revealed: simpler model achieves 91%, all paradigms share the same recall asymmetry, SOA beaten by 3 of 4 solutions. None of this is visible from a single solution.

### FUTURE WORK

Threshold optimization + cost-sensitive learning · External validation on Hungarian, Switzerland, Long Beach · Calibration to convert AUC strength into clinically reliable probabilities.

### TEAM LEARNING

The Comparative Multi-Solution Approach taught us that the most valuable insight is rarely which model wins — it is WHY a paradigm wins or loses, and what that reveals about the problem structure and data itself.

# Questions?

*Thank you.*

## REFERENCE SUMMARY · ALL FIVE MODELS

| Solution                 | Accuracy | Precision | Recall | F1     | AUC-ROC |
|--------------------------|----------|-----------|--------|--------|---------|
| Sol #1 Neural Network    | 85.00%   | 95.24%    | 71.43% | 81.63% | 0.9297  |
| Sol #2 Logistic Regress. | 91.00%   | 100.00%   | 82.00% | 90.00% | 0.9500  |
| Sol #3 K-Means + SVM     | 91.67%   | 100.00%   | 82.14% | 90.20% | 0.9319  |
| Sol #4 Naïve Bayes       | 86.70%   | 100.00%   | 71.00% | 83.00% | 0.9100  |
| SOA Random Forest        | 70.00%   | 66.67%    | 71.43% | 68.97% | 0.8500  |



 CITATIONS

# Sources cited in this presentation.

[1] R. Detrano, A. Jánosi et al., "International Application of a New Probability Algorithm for the Diagnosis of Coronary Artery Disease," Am. J. Cardiology, vol. 64, no. 5, pp. 304–310, 1989.

[2] D. Dua and C. Graff, "UCI Machine Learning Repository," University of California, Irvine, 2019. Available: <https://archive.ics.uci.edu/ml>

[3] S. Pouriyeh, S. Vahid et al., "Comprehensive Investigation and Comparison of Machine Learning Techniques in the Domain of Heart Disease," IEEE ISCC, 2017.

[4] M. A. Rahman, "Prediction of Heart Disease Based on Logistic Regression and Random Forest Models," ResearchGate, 2021.

[5] P. Sharma and A. Gupta, "Early Heart Disease Prediction Using Feature Engineering," ScienceDirect, 2022.

[6] K. Rani and R. Bhatt, "Prediction of Heart Disease UCI Dataset Using ML Algorithms," ResearchGate, 2020.

[7] D. Lapp, "Machine Learning for Heart Disease Prediction," Scribd.

